Module 08: Planning — Coding a Game

Learning Objectives

1. Understand that **Planning** is the first step of coding.
2. Learn about **Logic** (If This, Then That).
3. Know that you can be a **Creator**, not just a Consumer.

Introduction: The Architect

Before you build a house, you need a blueprint. Before you write code, you need a plan. What is the game about? Who is the hero? How do you win? Planning is imagining the future and working backwards.

Key Concepts

1. The Goal (The Objective)

Every game needs a goal.

- Save the Princess.
- Collect 100 coins.
- Survive the zombie apocalypse. Without a goal, it's just a toy, not a game.

2. The Logic (If/Then)

Computers think in logic.

- **IF** Mario touches a mushroom, **THEN** he gets big.
- **IF** Mario touches a spike, **THEN** he shrinks. This is the "Physics" of the digital world.

3. Loop (Repeat)

Computers are great at doing the same thing over and over.

- "Spawn a new enemy every 10 seconds."
- "Keep playing music until Game Over." Loops make the game run.

Activities

Activity 1: Design a Level

Draw a level for a platformer game (like Mario).

- Where is the start?
- Where is the goal?
- Where are the enemies?
- Is it too hard? Too easy? (Playtesting).

Activity 2: "If This, Then That" (IRL)

Make rules for your house using code logic.

- "IF it is 8:00 PM, THEN brush teeth."
- "IF room is messy, THEN no iPad."
- "IF dinner is eaten, THEN get dessert." Coding is just rules for life!

Summary

Don't just play games. Make them. The world needs your ideas.

References

- *Scratch* (Programming for Kids) by MIT
- *Code.org* (Hour of Code)